

# The Scientific Method

A philosophical framework for doing science. Objective and free from bias and prejudice. Objective: "Not influenced by personal feelings or opinions in considering and representing facts" from Oxford English dictionary

An iterative and systematic approach, collecting evidence from experiments, empirically testing our beliefs against data, and aiming for valid conclusions and reasonable explanations.

## 1 Scientific Inference

Inference is a conclusion reached on the basis of evidence and reasoning. Scientific inference is the process of reaching conclusions based on available evidence and theories. We have one or more premise(s), from which we draw a conclusion.

Can be split into Deductive inference and Inductive inference

### 1.1 Deductive Inference

An inference is deductive if there is a relation between the premises and the conclusion that guarantees the truth of the conclusion if the premises are true. If the premises are true, the conclusion must be true.

### 1.2 Inductive Inference

also known as induction, an inference is inductive if we use premises about some observed object to draw conclusions about similar unobserved objects. Implies that true premises can yield a false conclusion.

We mostly use inductive inference in research, we cannot test every possible input combination, we cannot evaluate with every person, etc. Thus, we observe a sample to draw conclusions about a population.

David Hume's problem of induction: When we use inductive inference, we assume the uniformity of nature, we assume unobserved objects are the same as the observed objects. Hume argues induction cannot be rationally justified.

### 1.3 Inference to the Best Explanation

Inference to the best explanation: Drawing the most plausible conclusion from the observations, similar to inductive inference (both are non-deductive)

How do we determine the best explanation? Occam's Razor "*More things should not be used than are necessary*" I.e. the most likely explanation for an occurrence is the simplest, requiring the least speculation.

### 1.4 Causal Inference

A common research goal is to explain what causes something, for example, why is text entry poor on touchscreens. Causal inference is using inference to reason about the cause of something. We can detect that something has happened, but not directly observe it actually happening.

Correlation does not imply causation, observing a correlation between X and Y does not mean that X causes Y. If text entry is poor on touchscreen keyboards, we cannot assume that touchscreen keyboards cause poor text entry.

We safely make causal inferences with controlled experiment design. There may be factors that affect our variable of

interest, not just the factor we are most interested in. We can control the effects of other factors through experiment design.

By controlling the effects of other factors through experiment design, we attempt to isolate the effect of our factor of interest, i.e. we compare our treatment group to the control group. If there's a difference between treatment and control, we can more safely infer that treatment is the cause of the outcome.

Even if we try to control every part of an experiment, there may be more factors, for example, taking the previous example, the treatment group might have inexperienced touchscreen users, people with larger fingers, etc. These are confounding factors.

Solutions to this? Randomised control experiments, where participants are put in groups at random, and nobody knows what group someone is in. or Repeated-measures, where participants experience all conditions, if possible.

## 2 Scientific Hypotheses

A hypothesis is a specific and testable explanation for a phenomena. It should be specific, predictive, testable, and parsimonious. Particularly, it should explain what you expect to happen in a certain circumstance, make a prediction that can be empirically tested, be falsifiable, and be as simple as feasible.

A hypothesis is an attempt to explain an observation that cannot be described. An informed guess about the cause of the phenomena. Allows predictions about unobserved entities. Predictions are tested against empirical data from experiments.

Hypotheses and Theories are not the same, a hypothesis is a testable prediction, a theory is a hypothesis proven by testing. Future testing may prove hypotheses to be true or false. The scientific method is an iterative process, hypotheses and theories adapt to new evidence.

## 3 Experiments

Experiments gather empirical data to test hypotheses against. Recall from earlier that if we want to make valid use of inference, we need to use carefully controlled experiments.

Experiment designs vary in CS, for example HCI experiments might measure the time taken to complete a task, IR experiments might measure the performance of search algorithms with metrics like time and relevance, Computer Architecture experiments might measure the number of clock cycles necessary to complete a set of instructions.

All serve the same purpose, gathering data against which we can test our hypothesis.

We use experimental data to test our hypothesis, thinking back to inductive inference, we cannot gather data about every entity, but we can observe a representative sample to draw conclusions about an unobserved population.

Inferential statistical analysis often used in test hypotheses. Did variance in data occur by chance or because of our experiment.

# Experiment Design I

## 1 Hypotheses

As previously discussed, a hypothesis is a specific and falsifiable statement explaining a phenomena. A well informed prediction of what will happen, makes predictions about unobserved entities. It should be testable against empirical data.

A hypothesis predicts relationship between independent and dependent variables, independent are controlled factors, i.e. conditions of observation, and dependent are measurable outcomes, i.e. experimental data.

For example, we could have the hypothesis *the super search algorithm will improve perceived relevance ratings of search results*. Independent variable here is ranking algorithm, super search rank vs baseline. Dependent variable is the document relevancy judgements.

This hypothesis is specific- it clearly identifies the variables of interest, it is predictive- clearly identifying an anticipated outcome, and it is falsifiable- we can design an experiment to test.

## 2 Experiment Design

From a hypothesis, we make experiment design. Experiment design identifies how data will be collected to test the predictions in a way that is fair, generalisable, and valid.

### 2.1 Independent Variables

Factors controlled or manipulated by the experimenter. Often related to the thing you have designed, e.g. your new algorithm, new interaction technique, new system, etc.

Usually categorical variables with different levels, e.g. algo 1 vs algo 2. Determines experiment *conditions* (all level combinations), for example, (A, low), (A, med), (A, high), (B, low), (B, med), (B, high)

### 2.2 Dependent Variables

Measurable values that are dependent on the conditions, can be measured or observed by the experimenter. Dependency implies a possible causal relationship, testing if independent variables cause outcomes in dependent variables. For example, did super search rank cause outcomes in dependent variables. We use inferential statistical analysis to test this.

### 2.3 Confounding Factors

A hypothesis identifies the independent and dependent variables, we design an experiment to fairly test these predictions. Well designed experiments enable fair comparison. We want to isolate factors of interest, control helps minimise confounding factors, which might affect dependent variables

Confounding factors might lead to incorrect conclusions, e.g., did super search cause improved relevance, **or** did the users issue less complex queries than for the baseline task?

#### 2.3.1 Identifying Confounds

To take an example hypothesis : *the super sort algorithm will be faster than quicksort*

We need to design a fair experiment to compare these, but what would an unfair comparison look like?

- Implemented in a different programming language

- Executing on a different computer
- Giving each algorithm a different set of inputs
- Having test inputs which favour one algorithm
- Running one while doing a resource-intensive task
- Executing one in power-saving mode

These are all potential confounds – how do we mitigate these?

- implementing in the same programming language
- Run on the same machine under the same conditions
- Use a large set of input data
- Give each algorithm the same set of inputs

Programming language, machine & conditions, input data are all control variables

### 2.4 Controlled Experiment Design

1. Identify potential confounding factors, we can increase fairness by controlling these
2. Identify ways of mitigating confounding effects, by minimising their effects, we reduce their impact on the data

Some things cannot be anticipated or controlled, and these are random variables. Their values vary randomly, or in unpredictable ways, e.g people, different abilities, intelligence, motor control.

## 3 Generalisation

An experiment needs to lead to generalisable findings, conclusions should be widely applicable - e.g. is super search rank generally better or only in this experiment.

Generalisation is necessary for inductive inference, we can't evaluate a new input device with every person alive, we can't evaluate search algorithm with every word combination.

Looking again at the super sort hypothesis, *the super sort algorithm will be faster than quicksort*, this makes a general claim. We can't test the infinite set of inputs to these algorithms, so instead we pick a **sample** of inputs. There may be instances where our algorithm is faster, or some where its slower.

For this hypothesis to be accepted and useful, we need to show that our algorithm is generally faster than quicksort.

### 3.1 Sample Selection

- **Subject** - Entities in an experiment, e.g. people, images
- **Population** - The set of all subjects
- **Sample** - the set of subjects *in the experiment*

Generalisation requires a representative sample, inductive inference requires that we observe entities to extrapolate findings to 'like' entities, i.e. the sample is representative of the population.

#### 3.1.1 Examples

CV uses sets of images to draw general conclusions about how recognition algorithms perform on *most images*. What is important is that these images are representative of the intended use case, e.g. are the images like those that would be captured by self-driving car sensors?

HCI uses groups of participants to draw general conclusions about how most people will interact with technology. Similarly to the previous example, are these people representative of the relevant or intended population?

## 3.2 Causal Inference

A fair and generalisable experiment supports causal inference. We want to show that the IV likely caused a change in the DV. We want to show this causal relationship is generally true.

Suppose we run supersort and quicksort 5000 times each on 200 randomly generated inputs sets. Lets say super sort was on average of 10% faster than quicksort. Did supersort cause the faster execution times?

We identified and attempted to control confounding factors, we repeated the measurements and varied the inputs to support generalisation. We **can** argue that it is probable that the IV (the algo) *caused* the difference.

## 3.3 Experiment Designs

Can be either within subjects or between subjects

**Within subjects design** - Subjects are exposed to all conditions, i.e. every level of the independent variable(s). Also known as repeated measures design, because you repeat the observations for every condition.

**Between subjects design** - Subjects are placed into groups and exposed to just one condition, i.e. they experience one level of the independent variable(s). Also known as between-group or independent measure designs, because you make independent measurements for each condition.

Within subjects designs are less common in other fields, like medicine for example, as the participants cant always take the treatment and control drugs at the same time or in sequence. Between subjects designs are typically used in other fields because the subjects can only be exposed to one of the conditions

In CS, within subjects designs are more common, digital systems can be returned to their initial state, we can process the same image with multiple vision algorithms. Human subjects tend to be minimally affected by condition exposure, they have limited lasting effects from one condition.

### 3.3.1 Within Subjects Advantages

Within subject experiments are sometime fairer, reducing confounding effects of subject characteristics, e.g people with better spatial skills who benefit one condition. They also tend to be cheaper and easier. If recruiting human participant, everyone experiences n conditions, instead of having to hire n times as many people.

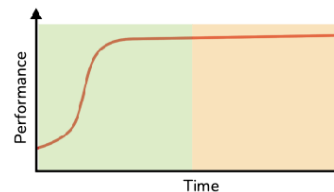
### 3.3.2 Within Subjects Disadvantages

Order effects are a potential limitation of within subjects designs, when the order of conditions has a potential impact on the DVs. Most common in experiments with human participants.

Learning effects are a particular type of order effect, when participants gain knowledge and experience that benefits the conditions towards the end of the experiment. For example, participants might achieve higher performance later on in an experiment because they learned how the UI works.

## 3.4 Order Effects

Learning occurs during the **first condition**, so performance will naturally be better in the **second condition**. The variance in our dependent variable is more likely to be caused by the learning effects than differences in the independent variable(s)



Randomising condition order might mitigate order effects. Conditions are presented in a random order for each participants, and so order effects get distributed across conditions in an attempt to cancel out the effect. This is less idea for small samples and lots of conditions.

Counterbalanced designs are an even better alternative, instead of hoping the random order cancels out learning effects, counterbalanced designs are systematically balanced. It would be impractical to use every unique condition ordering ( $n!$  orderings).

Instead, we use a Latin square, an  $n \cdot n$  array where each symbol occurs only once in each row and once in each column. This is a practical compromise, yields n condition orders such that each condition will appear just once in each position in the sequence

## 3.5 Latin Square

An easy way of creating a counterbalanced experiment design, if order effects need to be controlled for

| Order 1 | Order 2 | Order 3 | Order 4 |
|---------|---------|---------|---------|
| A       | B       | C       | D       |
| B       | C       | D       | A       |
| C       | D       | A       | B       |
| D       | A       | B       | C       |

# Experiment Design II

A good hypothesis identifies experiment conditions, necessary observable outcomes, and makes falsifiable predictions that you can test. Experiment design identifies *how* data will be collected to test the predictions in a way that is fair, generalisable, and valid.

## 1 Experimental Validity

Does the experiment **credibly** achieve its aims?

Experiments often take place in controlled environments, which makes it easier to mitigate potential confounding or distracting factors. Experiments aim to generalise to a variety of authentic scenarios (is our controlled environment representative of the chaotic world?)

Not all designs lead to correct and applicable conclusions.

### Internal Validity

- Is it valid to infer a causal relationship between the independent and dependent variables?
- Does the experiment design support the conclusion?

### External Validity

- Is it valid to generalise findings from this experiment to another context or population?
- Would the finding generalise to a different sample?
- Would the finding generalise to a different task?

### Ecological Validity

- Is it valid to extrapolate the findings to authentic application contexts?

- Does this experiment reasonably represent the real world?

What is difference between external and ecological validity? Well external is concerned with whether the finding applies to *other* contexts, and ecological is concerned with whether the finding applies to *realistic* contexts. Ecological validity is considered a subset of external validity - *does it generalise to reality?*

## 1.1 Experiment Environment

We need to make a trade-off between control and ecological validity, i.e. variability and unpredictability vs authenticity. Consider what is most important for the experiment *at this time*. Could be formative studies in a lab, and comprehensive evaluations in the wild.

Controlled studies can try to replicate authentic scenarios.

- Can the setting be made more realistic?
- Is the prototype system authentic?
- Is the task comparable to a realistic task?
- Do the participants face similar demands to a genuine scenario?

## 1.2 Experiment Tasks

An experiment task is an instance of the event being observed, e.g. an object detection algorithm processing an image, a user completing a sequence of taps, etc. At minimum there needs to be one task per condition, so that dependent variables can be measured at least once per condition.

Most experiments have multiple tasks per condition, taking the average across n tasks reduced effect of outliers for reliable data. Allows for natural variability across task instances.

### 1.2.1 Task Validity

Consider the internal, external, and ecological validity of the **task**. Does data from this task address the research question? Do findings from this task tell us about other tasks? Could this task tell us something about realistic use cases?

### 1.2.2 Task Abstraction

Sometimes experiment control necessitates abstraction, to manipulate independent variables in an abstract task or to limit confounding factors. Abstract tasks help generalise, e.g. systematically investigate input performance in a generic interface to extrapolate to any user interface design.

## 1.3 Sample Validity

David Hume's problem of inductivity. Is there uniformity of nature? i.e. is the unobserved 'the same as' observed?

Is the experiment sample valid? is is representative of the unobserved? is it representative of practice? respectively, these are external and ecological validity.

A sample needs to be representative of the broader population. If the sample is not representative, we can't draw general conclusions. Does the sample have similar character-



istics as the population? does your sample have internal & external validity?

### 1.3.1 Sample Validity Examples

Trade off between validity and practical concerns:

- Getting access to more relevant participants may be difficult
- Safeguarding potentially vulnerable participants

Consider the current research progress, is the research mature enough to test with a 'more valid' sample? are you making good use of peoples' time?

## 2 Sample Size

How large should a sample size be? Sample size has implications for representation, for example, too homogenous a sample lacks variability.

Sample size also has implications for inferential statistical analysis. Lack of statistical power, and so failing to recognise an existing effect. Known as a type II error, or false negative. We use experiments to test our hypotheses, but statistical power implies possibility of incorrect conclusion

### 2.1 Statistical Power

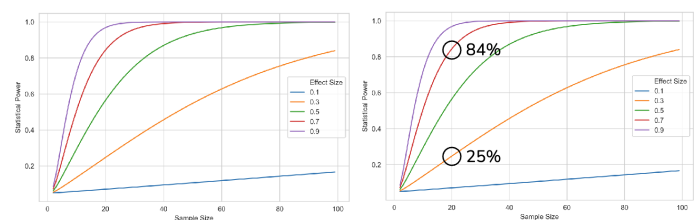
Statistical power is the probability of detecting an existing effect. As statistical power increases, the probability of a type II error decreases. In other words, Low power implies increased probability of failing to detect the effect, and high power implies increased probability of detecting the effect.

Power depends on the statistical test, sample size, and the data. A power analysis helps estimate the necessary sample size. In practice, the ideal sample size needs balanced against practicality.

#### 2.1.1 An Example

Following example is for a t-test, testing if two conditions have significantly different mean values. E.g. comparing task completion time for two different input techniques.

Simulated experiment with up to 100 subjects, comparing data from two conditions using a t-test, effect sizes ranging from 0.1(weak) to 0.9(strong), this is Cohens effect size of t-test - ratio of standard deviations between conditions



### 2.2 Statistical Power(cont.)

Statistical power is determined by both factors you can and cannot control. Both by experiment design and sample size, and by effect size.

Key point is that a larger sample size increases statistical power and increases the likelihood that your sample is representative. Again, balance sample size against practical concerns - how many participants can you hire, how much time do you have, etc... For digital 'subjects', prepare the largest practical sample, e.g. images, tweets, graphs, arrays, traces, etc.

## 2.3 Local Standards

When using human participants, recruit as many as is practical, ideally at least 15-20 people. This is an arbitrary target that likely falls short of power analysis targets. Aim to be consistent with local standards in the field, i.e. how many people typically participate in published studies like yours?

# Quantitative Analysis I

## 1 Experiment Data

Experiments try to falsify predictions made by hypotheses, i.e testing relationship between independent and dependent variables. Hypothesis testing determines if evidence supports predictions, is the data consistent with what was expected?

Hypotheses are accepted or rejected based on strength of evidence, is there sufficient evidence to suggest the hypothesis was correct? Using inferential statistical analysis to empirically inform our decision

### 1.1 Experiment Data Types

Quantitative data is data that expresses a numerical quantity, for example, compilation time, classification accuracy, completion time, task success. Qualitative data is *non-numerical*, for example, interview transcripts, observation notes, videos.

Most scientific research data is quantitative, but qualitative is common in some areas, e.g. HCI, CS education, and SE.

### 1.2 Data Sources

Experimental data can be generated in real time in code, e.g. measuring algorithm execution time or task completion time. It can be generated from other data, e.g. metrics calculated from motion data traces, processing the outputs of an algorithm. Data can also be generated by humans, i.e. numerical survey ratings.

### 1.3 Data Mistakes

Dependent variables are poorly defined or lack validity. Are you measuring what you intended to? for example, how do you measure effort?

Inconsistent or non-repeatable measurements, is the data being accurately and reliably measured? e.g. instrumented software vs stopwatch

Incorrectly formatted or measured. Does the data have appropriate precision and resolution, e.g. measuring time in seconds instead of milliseconds

### 1.4 Pilot Data Collection

Use pilot experiments to test data collection, test things like

- Does the observation methodology work?
- Can you access the data?
- Is it accurate and usable?
- Does it work with your analysis approach?

### 1.5 Data Formatting

Data needs to be in a format that can be easily processed, most likely in a tabular format with consistent structure. We

| User | Block | Task | IV1(Device) | IV2 (Feed-back) | DV1 (Task Time) | DV2 (#Mistakes) |
|------|-------|------|-------------|-----------------|-----------------|-----------------|
| 1    | 1     | 1    | Touchscreen | Visual Only     | 3105            | 5               |
| 1    | 2     | 1    | Touchscreen | Visual + Audio  | 2897            | 3               |
| 1    | 3     | 1    | Mouse       | Visual Only     | 5470            | 3               |
| 1    | 4     | 1    | Mouse       | Visual + Audio  | 5122            | 2               |
| 1    | 1     | 2    | Touchscreen | Visual Only     | 3299            | 4               |
| 1    | 2     | 2    | Touchscreen | Visual + Audio  | 2909            | 0               |
| 1    | 3     | 2    | Mouse       | Visual Only     | 6121            | 2               |

need to know what the data is describing, including the relevant meta data, e.g. independent variables, random variables. We want as much descriptive data as possible.

We should be aware of dependent variable integrity and precision, e.g. using correct units with sufficient points after decimal

## 2 Inference

Inference is reaching conclusions through evidence and reasoning. Inductive inference is drawing conclusions about unobserved entities based on evidence from observed entities (We cannot assume the uniformity of nature). Inference to the best explanation is reaching the most plausible explanation from available explanation. Think back to Occams razor. Causal inference is inferring what causes something to occur, remember that correlation does not imply causation, and controlled experiments attempt to isolate the factor we're interested in.

### 2.1 Estimation

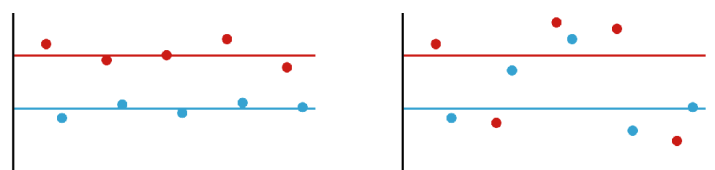
Estimating dependent variable from a sample of the population. That is, a sample of size  $N$  gives a mean  $\bar{x}$  that estimates the population mean  $\mu$ . As  $N$  increases,  $\bar{x}$  approaches  $\mu$  - it becomes a more accurate estimation. Recall sample size is also important for generalisation and statistical power.



Experiments collect data from a sample of a larger population. Your experiments is just one of many that could have occurred. Similarly, Your sample is just one of many you could have investigated.

Most experiments give an estimate near the true value, but we don't objectively know the true value. It's possible that our estimate is poor but we don't know.

#### 2.1.1 Descriptive Statistics?



Descriptive statistics can be misleading, the same two mean values can come from different distributions of points. How do we decide if there's a true difference between red and blue



## 3 Inferential Statistical Analysis

Statistical tests that support the use of inference. Did the independent variables cause an effect on dependent variables? What degree of confidence do we have in our estimates? Can we make predictions about the population from this sample?

### 3.1 Null-Hypothesis Significance Testing

A widely used approach for inferential statistical analysis, not the only one though, there are Bayesian alternatives. This is based on the probability of obtaining **these** values. Terminology for NHST concepts is widely used to discuss research results, things like statistical significance, p-values, significant differences, etc.

Attempt to falsify the null hypothesis with your data. A hypothesis can be stated as a null and alternate hypothesis. The null-hypothesis,  $H_0$  states there is no (meaningful) effect or relationship between independent and dependent variables. The alternative  $H_A$  states there **is** an effect or relationship, i.e. a causal relationship. If we reject the null hypothesis, we accept the alternative

#### 3.1.1 Example

Q: Is speech faster than touch for button activation?  $H_A$  is that speech input will lead to faster activation than touchscreen input. We can design an experiment to test this hypothesis. Our IV is input (speech or touchscreen), DV is task completion time, and the task is to complete a sequence of button activations.

Suppose we ran this experiment with 20 people, and speech input took a mean of 3050ms, and touch input took an input of 4200ms. Is this 1150ms difference statistically meaningful? We'll use a within-subject t-test.

Estimated probability implies if we should reject null hypothesis, we can **falsify** the null hypothesis, but cannot **prove** the alternative hypothesis. If we reject  $H_0$ , we consequently accept  $H_A$

NHST estimates the probability of an observation if  $H_0$  was true, i.e. if there were no effect, how probable was this data. This probability  $p$  is the p-value. If  $p=0.45$  then there is a 45% we could obtain this observation if  $H_0$  was true.

P values are compared to a predetermined threshold  $\alpha$ . If the p-value is below this threshold, we reason that the probability of obtaining this result is too low for the null hypothesis to be true. It is highly improbable that we would obtain this result if  $H_0$  was true. Thus, if  $p < \alpha$  we reject the null hypothesis

The significance threshold is  $\alpha$ . Common values of  $\alpha$  tend to be 0.05 and 0.01. If  $p$  is less than alpha, we take the result to be statistically significant. If we were comparing two values, we could say there was a statically significant difference between them.

Now, taking the earlier example, if we take  $\alpha = 0.05, p = 0.03$ , we have evidence to reject  $H_0$  and consequently accept  $H_A$ . The 1150 ms difference would be highly unlikely if there was no effect,

**A Caveat:** We're making an inference using an arbitrary threshold, still a non-zero chance we obtain this result if  $H_0$  was true. Also, our data gives one of infinite estimations, our estimation might be inaccurate - bias could work for or against us.

There are limitations to using this approach, inductive inference can lead to incorrect conclusions, but if experiment

is valid and probability suggests it is *probably* true, we can reason that it likely is.

## Quantitative Analysis II

### 1 Statistical Tests

Remember, We cannot prove a hypothesis is true, but we can falsify a hypothesis. Null hypothesis significance testing is based on this principle. If we can falsify the null hypothesis of no effect, then the evidence supports the alternative hypothesis – that there is an effect.

Many statistical tests can be used for significance testing, choose this based on your experiment design and data characteristics. Statistical functions calculate a test statistic from the data,

- t-test calculates the t statistic, the ratio of difference to variability
- ANOVA calculates the F statistic, ratio of variance across groups
- Wilcoxon test calculates the W statistic, sum of ordinal ranks

Probability is determined by the test statistic distribution, giving you the corresponding p value for this samples result.

Each statistical test has a null and alternative hypothesis. You use the p value to attempt to falsify the null hypothesis.

#### Repeated Measures t-test:

- $H_0$ : No difference between *two* group means
- $H_A$ : Significant difference between *two* group means

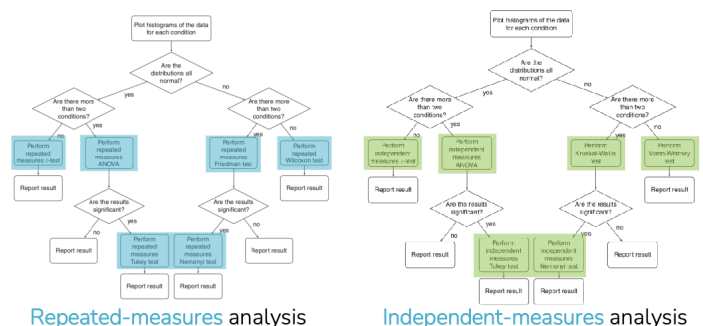
Similarly, **Repeated-measures ANOVA:**

- $H_0$ : No difference between  $k$  group means
- $H_A$ : Significant difference between  $> 2$  group means

#### 1.1 Choosing a Test

Statistical tests are just functions, in that they assume you know what you're doing. The t-test wont suggest you use a Wilcoxon signed-rank test instead. You might also use the correct test, but incorrectly. For example, forgetting to aggregate repeated measures, not handling outliers, or incorrectly specifying model parameters.

It is important to know how to choose and use tests correctly. Not as necessary to know *how* the statistical tests work, but you need to justify that your methods are appropriate. Some statistical tests make assumptions about the data. Be aware of these and whether your data meets them. For example, many tests assume a normal distribution.



## 1.2 Parametric vs Non-Parametric

Parametric tests make assumptions about data distribution, e.g. assuming data is from a normal or Poisson distribution. Non-parametric tests **do not** make assumptions about data, it is often ranks, then the ranks are analysed.

Most parametric tests have a non-parametric equivalent, e.g. t-test and Wilcoxon signed-rank, ANOVA and Friedman's

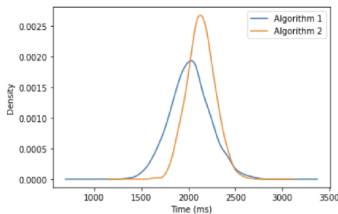
Parametric tests are typically used for continuous data, e.g. processing time, f1 score, task completion time. There is a lack of consensus on how to handle ordinal data. Survey scales might not meet some parametric assumptions.

## 2 Example Analysis

Comparing two algorithms to see if one is faster

- **Independent variable:** Algorithm
- **Dependent variable:** Processing time
- **Sample:** 1000 sets of input data
- **Task:** running the algorithm with given input data
- **Analysis:** repeated-measure t-test

Our hypotheses are  $H_0$  there is **no** difference between mean processing time and  $H_A$  there **is** a significant difference between the mean processing time. Summary statistics show



|      | Algorithm 1 | Algorithm 2 |
|------|-------------|-------------|
| mean | 2019.462084 | 2134.738268 |
| std  | 213.140030  | 148.589010  |

mean times are 2019 and 2135ms. This is not convincing evidence of a meaningful difference. An exploratory plot shows the distribution of execution time, showing that Algorithm 1 looks marginally faster. Again, this is not convincing that there is a meaningful difference.

$$t = -14.06, p = 4.214113672159636e - 41$$

With a p-value less than 0.001, we can reject  $H_0$ , the data supports  $H_A$ , the mean times **are** significantly different.

### 2.1 Reporting Results

When it comes to reporting results, we want to give as much info as we can, including the statistical test used, F-statistic, degrees of freedom, p-value, and effect size.

A repeated-measures ANOVA found a significant effect of user interface design:  $F(2,38) = 5.3, p = .009, \mu_p^2 = .22$

#### 2.1.1 Post-Hoc Procedure

Many group-tests like ANOVA tell us if there is a difference, but do not identify the difference. If there is a significant difference, perform post-hoc pairwise comparisons. Use p-value correction for multiple comparisons if appropriate. If there is no significant difference, **do not** do post-hoc tests.

#### 2.1.2 Factorial Analysis

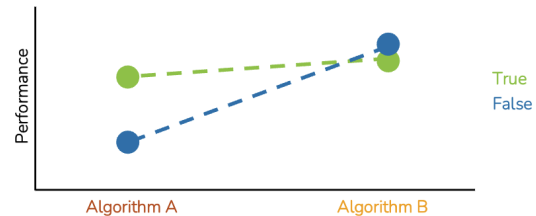
Our example had a single independent variable - two conditions. Many experiments have multiple independent variables, for example, image recognition algorithm (A,B) x preprocessing (True, False). In these cases, we want to investigate if the *algorithm* affects recognition performance, or if

*image pre-processing* affects recognition performance. Use factorial analysis to investigate if *each factor* has an effect, and to investigate if there is an interaction effect between factors.

#### 2.1.3 Interaction Effects

Be aware of interaction effects between factors, we could use ANOVA to investigate the effect of each factor. Running the test gives us three results;

- the *main effect* of algorithm (A,B) on DV
- the *main effect* of preprocessing (true,false) on our DV
- the *interaction effect* between algorithm and preprocessing



Algorithm B might only be better than Algorithm A when preprocessing is false.

## 3 Poor Statistical Practice

There is a desire to avoid "negative results", e.g. where  $p > 0.05$ . There is a positive outcome bias, so it might be more difficult to publish.

In pursuit of a positive result, we might fall into some bad practice, e.g. manipulating or fabricating data, misusing or misinterpreting p-values, or HARKing (Hypothesising After Results are Known) to change hypothesis to suit the data.

To combat poor statistical practice, we should

- Drive towards improved statistical transparency - making it easier for others to check your work
- Registered reports, pre-registering experiments and analysis - peer review papers before the experiment, reducing the desire to abuse analysis to get preferential results
- Increased Scrutiny of published results

## Qualitative Research

Investigates the more 'human' aspects of a research topic, i.e. things that cannot necessarily be quantified, e.g. behaviours, opinions, emotions, experiences, social factors, etc.

Involves collection and analysis of non-numerical data and artefacts, like interview transcripts, open ended survey responses, photos and videos, co-created user interface designs.

## 1 Qualitative Computing Science

Software engineering is an example, we use ethnographic observation of teams to understanding working practices, or interview software developers to get their perspectives, or analyse their use of version control.

Another field which makes use of qualitative research is HCI, we might interview participants to understand mental models, observe users to see how they do things, or have focus group discussions with domain experts to get requirements.

Similarly, CSE is another example, we might interview students about their understanding of a topic, or observe teachers to identify aspects of their teaching practice, or have focus groups to discuss student reflection and revision habits.

Qualitative research is inherently subjective. Researchers are actively in the research context, with their own beliefs, opinions, perspectives, and values. They may have prejudices that affect how they see the research problem.

Qualitative research can be subject to bias, like confirmation bias (favouring an interpretation that fits hypothesis) or reflexivity bias (favouring interpretation that fits beliefs, culture, etc.)

Qualitative bias might not necessarily be an issue, researchers being actively engaged can be a good thing, they might have connections with participants, empathy, etc.

While quantitative research is a more objective process, following systematic processes and following rules, its still subject to bias and misuse, the researchers are still **interpreting** the results.

## 2 Mixed Method Research

Using both qualitative and quantitative is *mixed methods* research, benefitting from the relative strengths of each approach. A combination can give richer insights than either method alone

### 2.0.1 Validation

Can validate findings with multiple kinds of evidence. Corroborate quantitative results with supporting qualitative data, for example, interview comments that support measured data. Mixed methods can increased both internal and ecological validity, by using qualitative to support quantitative results and demonstrating real-world context of results.

### 2.0.2 Explanation

Qualitative insights can help explain quantitative results, that is, participant perspective might help us understand *why* input technique A had faster task completion time than input technique B

### 2.0.3 Perspectives

Qualitative gives a different perspective on quantitative results, e.g. two input methods had similar task time, but users disliked one because they had to focus more on what they were doing. Not everything can be quantified - you might not have known *what* to quantify in the first place. Qualitative methods can help uncover the bigger picture.

## 2.1 Qualitative Data Collection

Qualitative data comes in many forms, and from many places, but is predominantly text-based. Commonly used data collection methods are survey responses, interview or focus group discussions, or observation notes.

### 2.1.1 Survey Design

Ensure questions measure the concepts you intend to measure - could answers to this tell me what i want it to?

Ensure questions are not ambiguous - will all participants know what this means?

Ensure questions are neutral and non-leading - could this question bias participants towards a certain response?

### 2.1.2 Interview Design

Plan an interview structure with important and relevant questions - if we only discuss these, we can at least answer the research question, use this structure to get back on track.

Be reflexive to what participants are saying - deviate from the structure to pursue interesting comments, in a semi-structured interview style. If unanticipated but interesting topics come up, ask other participants in future too.

Ask neutral questions, do not make assumptions, avoid leading questions.

## 3 Qualitative Coding

How do we analyse predominantly text-based data, like from interview transcripts, survey responses, and observation notes?

Use qualitative coding - the process of assigning codes to portions of text data. Codes are labels that identify some kind of thematic categorisation.

Codes give a variety of insights - codes themselves identify common themes in the data, they quantify prevalence, and we can look at coincidence and interaction between codes - how often do two codes occur together.

### 3.1 Approaches

Coding can be **Deductive**, **Inductive**, or **In Vivo**.

- **Deductive Coding:** When you have already identified the codes before analysis begins
- **Inductive Coding:** When you identify codes during analysis. This is responsive and organic, need to be open to what the data says.
- **In Vivo Coding:** Inductive coding where codes are direct quotes.

Generally, Coding is an iterative process, especially for inductive approaches. Start with **Open coding**, the initial labelling of text, followed by **Axial coding**, structuring codes into broader themes, then finally **Selective coding**, focusing on the most important codes.

The process, generally speaking is,

- Formalise the approach, write the methods sections
- Familiarise yourself with the data, read the transcripts
- Open coding, go through and identify initial codes
- Axial coding, restructure codes into themes
- Selective coding, do another code pass in need be
- Analysis, look at prevalence, themes, review coded text
- Writeup, interpret the coded data, what it means.

### 3.2 Reliability

Reliability concerns the validity with which the codes are applied, are they being applied accurately and consistently? Coding review improves confidence in reliability. Reliability can be split into **Inter-** and **Intra-coder** reliability.

Inter-coder reliability refers to how consistently the codes are applied by multiple people, which is especially important if coding is distributed across the research team. Intra-coder reliability refers to how consistently codes are applied by one person, this can be assessed by repeat coding to see if codes are used consistently each time